Object Oriented Programming (CSEG1044)

Unit 01 – Lecture 04: Control Flow + Arrays + Strings

April 1, 2026

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Quick Links

Overview Key Topics Demo Summary

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Lecture Snapshot

- **Unit focus:** Introduction to OOPs and Java Fundamentals

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Lecture Snapshot

- **Unit focus:** Introduction to OOPs and Java Fundamentals
- **Course thread:** Use Control Flow + Arrays + Strings to make the first text-based version of Campus Resource Manager easier to reason about before the full object model arrives.

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Recall Before You Start

- **Types and operators**: Conditions, loop bounds, indices, and string logic depend on the previous lecture. Main reference: `Unit 01 / Lecture 03 slides/notes`.

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Recall Before You Start

- **Types and operators**: Conditions, loop bounds, indices, and string logic depend on the previous lecture. Main reference: `Unit 01 / Lecture 03 slides/notes`.
- **Program flow**: Control statements only make sense once you already know the default top-to-bottom execution order. Main reference: `Unit 01 / Lecture 02 slides/notes`.

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Learning Outcomes

- Use `if/else`, `switch`, and loops to control program flow.

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Learning Outcomes

- Use `if/else`, `switch`, and loops to control program flow.
- Create/traverse 1D and 2D arrays and solve simple array problems.

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Learning Outcomes

- Use `if/else`, `switch`, and loops to control program flow.
- Create/traverse 1D and 2D arrays and solve simple array problems.
- Explain `String` immutability and use `StringBuilder` when needed.

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed April 1, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed April 1, 2026).

Key Terms to Know

- **Selection**: choosing one path among many (`if`, `switch`).
- **Iteration**: repeating work until a condition changes (`for`, `while`).
- **Array**: fixed-size contiguous collection of same-type elements.
- **Index**: position in array (0-based in Java).
- **Immutable**: cannot be changed after creation (e.g., `String`).
- **StringBuilder**: mutable sequence of characters (efficient for repeated concatenation).

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Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.

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Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.
2. Study the main build in this order: *Program control statements (selection + loops)* → *Arrays (1D/2D), types of arrays, common patterns* → *Strings and* `StringBuilder`.

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1. Refresh the prerequisite bridge before reading new ideas in isolation.
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3. Trace the worked example and connect each line back to one concept frame.
4. Attempt the mini exercises before moving to the recap and exit check.

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Unit 01: Introduction to OOPs

- Lecture focus: Control Flow + Arrays + Strings

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Today: Roadmap

- Program control statements (selection + loops)

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- Program control statements (selection + loops)
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Today: Roadmap

- Program control statements (selection + loops)
- Arrays (1D/2D), types of arrays, common patterns
- Strings (immutability, common methods), ‘StringBuilder‘ (need-based)

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Control flow (quick)

- Selection: `if/else`, `switch`

Common pitfalls

- Off-by-one errors in loops
- Missing `break` in `switch` (unintended fall-through)

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Control flow (quick)

- Selection: `if/else`, `switch`
- Loops: `for`, `while`, `do-while`

Common pitfalls

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Control flow (quick)

- Selection: `if/else`, `switch`
- Loops: `for`, `while`, `do-while`
- Useful keywords: `break`, `continue`

Common pitfalls

- Off-by-one errors in loops
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Arrays (1D/2D)

- 1D: `int[] a = new int[n];` and `a.length`

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- Traversal: `for (int i=0; i<a.length; i++)` or `for (int x: a)`

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Arrays (1D/2D)

- 1D: `int[] a = new int[n];` and `a.length`
- Traversal: `for (int i=0; i<a.length; i++)` or `for (int x: a)`
- 2D: `int[][] m = new int[r][c];`
- Jagged arrays: each row can have different length (use `m[i].length`)

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Strings (must-know)

- `String` is **immutable**: methods return new strings.

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Strings (must-know)

- `String` is **immutable**: methods return new strings.
- Compare content with `equals()`, not ==.

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Strings (must-know)

- `String` is **immutable**: methods return new strings.
- Compare content with `equals()`, not `==`.
- Repeated concatenation in loops is slow → use `StringBuilder`.

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Checkpoint and Reflection

Reflection prompt

Where would *Control Flow + Arrays + Strings* matter most in Campus Resource Manager, and which design choice would you justify using this lecture's ideas?

Quick self-check

Which part needs one more example before you move on: *Program control statements (selection + loops)*, *Arrays (1D/2D), types of arrays, common patterns*, the worked example, or the recap?

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Example: grades (arrays + control flow) (1/2)

```java
import java.util.Scanner;

public class GradeStats {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of students: ");
        int n = sc.nextInt();

        int[] marks = new int[n];
        for (int i = 0; i < n; i++) {
            System.out.print("Marks for student " + (i + 1) + ":
                ");
            marks[i] = sc.nextInt();
        }
```

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Example: grades (arrays + control flow) (2/2)

```
// continued...
        int max = marks[0], sum = 0;
        for (int x : marks) { sum += x; if (x > max) max =
            x; }
        double avg = sum / (double) n;
        System.out.printf("Max=%d, Avg=%.2f%n", max, avg);
        sc.close();
    }
}
```

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Mini Exercises (2–3 mins)

1. Array of length 10: what is the last valid index?

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Mini Exercises (2–3 mins)

1. Array of length 10: what is the last valid index?
2. Which compares string content: `==` or `equals`?

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Mini Exercises (2–3 mins)

1. Array of length 10: what is the last valid index?
2. Which compares string content: `==` or `equals`?
3. Which loop runs at least once: `while` or `do-while`?

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Watch-outs

- Writing off-by-one loop boundaries.

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Watch-outs

- Writing off-by-one loop boundaries.
- Comparing String values with == instead of an equality method.

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Watch-outs

- Writing off-by-one loop boundaries.
- Comparing String values with == instead of an equality method.
- Treating String as mutable and expecting in-place updates.

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Key Takeaways

- Use selection + loops to express logic clearly (avoid off-by-one bugs).

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Key Takeaways

- Use selection + loops to express logic clearly (avoid off-by-one bugs).
- Arrays store many values of the same type; `length` guides safe iteration.

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Key Takeaways

- Use selection + loops to express logic clearly (avoid off-by-one bugs).
- Arrays store many values of the same type; `length` guides safe iteration.
- Strings are immutable; use `equals()` and `StringBuilder` when needed.

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Checkpoint Questions

1. How do you avoid off-by-one errors in array loops?

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Checkpoint Questions

1. How do you avoid off-by-one errors in array loops?
2. Why should we use `equals()` for strings?

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Checkpoint Questions

1. How do you avoid off-by-one errors in array loops?
2. Why should we use `equals()` for strings?
3. When is `StringBuilder` preferred over `String` concatenation?

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Next Study Steps

- Revisit the recall references if any older idea still feels weak.

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Next Study Steps

- Revisit the recall references if any older idea still feels weak.
- Rework the lecture example without looking at the notes first.
- Attempt the self-study practice and then answer the exit questions in full sentences.
- Apply the idea once inside Campus Resource Manager to make the concept stick.

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Next Unit Preview

Next: Unit 02 – Classes, Objects, and Encapsulation

- We will start writing proper classes with private fields and methods.

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Next Unit Preview

Next: Unit 02 – Classes, Objects, and Encapsulation

- We will start writing proper classes with private fields and methods.
- Bring one example of a real-world entity you want to model as a class.

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